

Dynamic Assignment of Digital Linear Sessions to Synthetic Audience Units

Context in one paragraph

A measurement platform already generates weighted synthetic linear-TV sessions for audience reporting. The next step is to incorporate measured digital linear usage from an external measurement partner. Digital usage is, opposed to TV usage, unweighted, meaning that every session is representing exactly one viewer. To map the external digital sessions into the existing infrastructure, a new approach needs to be developed. This should dynamically activate and deactivate internal synthetic viewers when enough unweighted digital sessions match a synthetic viewers weight. The solution must preserve weighting and demographic consistency and support KPI comparability across multiple aggregation levels.

1. Objective

Design and integrate logic that dynamically activates, deactivates, and reactivates synthetic viewer based on qualifying digital-linear session volumes. The logic must:

- Match digital sessions and synthetic viewer by demographic bucket
- Use representational weights consistently for activation thresholds and capacity limits
- Produce session-shaped internal output comparable to existing synthetic linear-TV sessions
- Be scalable to different levels and should not increase the current export procedure by more than 30 Minutes

2. Functional scope

2.1 Digital session bucketing

For every second of the broadcast day, all active digital-linear sessions are grouped into six demographic buckets. A digital session contributes 1 to each second it spans, regardless of the session length or the user/device behind it.

Bucket	Gender	Age band	Notes
1	Male	12-29	One active digital session = 1 viewer
2	Male	30-49	One active digital session = 1 viewer
3	Male	50+	One active digital session = 1 viewer
4	Female	12-29	One active digital session = 1 viewer
5	Female	30-49	One active digital session = 1 viewer
6	Female	50+	One active digital session = 1 viewer

Missing demographic attributes

If a digital session lacks gender, age, or both, the missing attributes are assigned through a randomized imputation process before the session is bucketed. The implementation should make this process reproducible for audit and backfill runs, for example by using a deterministic seed based on stable session identifiers and the broadcast day.

2.2 Synthetic unit pool preparation

- After all baseline synthetic sessions for a broadcast day have been generated, build the pool of synthetic viewer eligible for digital activation.
- Prefer synthetic viewer that have not yet been activated on the current broadcast day.
- Sort eligible viewer by ascending representational weight within each demographic bucket.
- Assign each synthetic viewer to the same six demographic buckets used for digital sessions.

2.3 Activation and deactivation thresholds

Activation and deactivation are evaluated independently within each demographic bucket.

Rule	Formula	Interpretation
Activation	active digital sessions $\geq \text{round}(\text{weight} \times 0.5)$	A synthetic viewer becomes active once matching digital volume reaches at least half of its weight.
Deactivation	active digital sessions $< \text{round}(\text{weight} \times 0.5)$	An active synthetic viewer is deactivated once matching digital volume drops below half of its weight.
Capacity	linked sessions $\leq \text{round}(\text{weight})$	The number of linked digital sessions may never exceed the unit capacity.

Example: A synthetic unit in bucket Male 12-29 with weight 10 has activation threshold $\text{round}(10 \times 0.5) = 5$ and capacity $\text{round}(10) = 10$.

Example timeline: activation and deactivation

Male 12-29, weight 10 -> threshold = 5, capacity = 10.

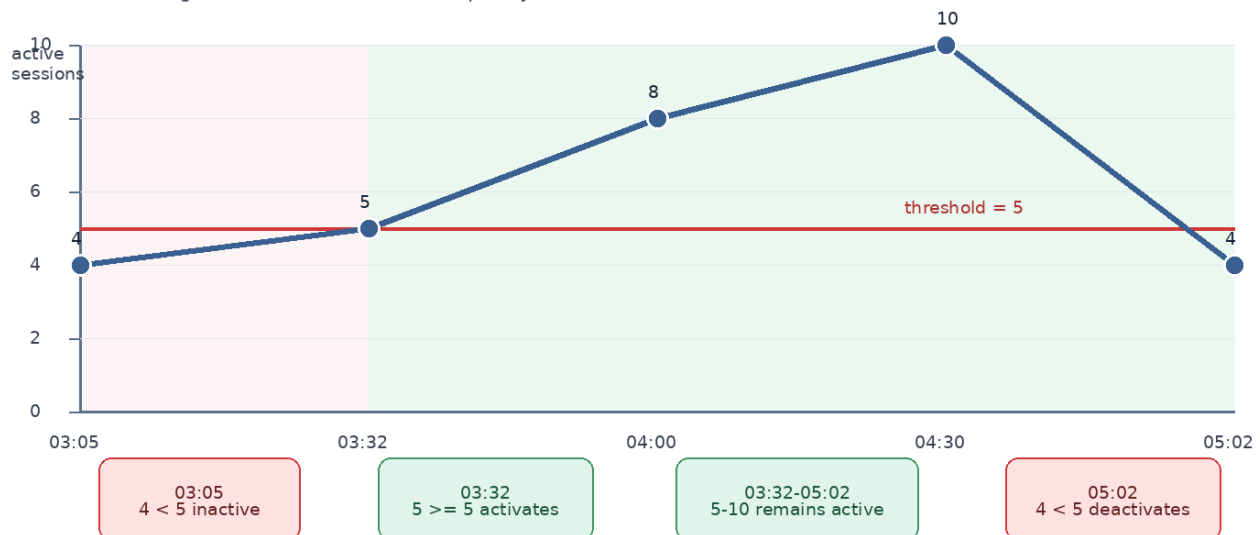


Figure 1: Simple activation/deactivation timeline for one demographic bucket and one synthetic unit.

3. Session-to-unit linking rules

3.1 Linking while active

- While a synthetic viewer is active, matching digital-linear sessions are logically linked to it.
- The total number of linked digital sessions must not exceed the rounded capacity of the viewer.

- Once the current viewer reaches capacity, additional matching digital sessions contribute toward activating the next eligible viewer in the same demographic bucket.
- The next viewer is chosen from the sorted eligible pool, starting with the smallest representational weight that satisfies the rules.

3.2 Deactivation and reactivation

- If volume drops below the 50% threshold, the active viewer is deactivated.
- A deactivated viewer remains eligible for reactivation later in the broadcast day.
- Viewer that have already accumulated digital sessions are prioritized on reactivation.
- Only after partially filled units are satisfied may empty or previously deactivated units be reused.

4. Illustrative scenario

Scenario setup: One baseline synthetic viewer exists for Male 12-29 with a representational weight of 10. Matching digital-linear sessions are observed over the broadcast day. The activation threshold is 5 and the capacity is 10.

Time	Matching digital sessions	Rule outcome	State
03:05	4	$4 < 5$, below threshold	Inactive
03:32	5	$5 \geq 5$, threshold met	Activates
03:32-05:02	5-10	Within threshold and capacity	Remains active
05:02	4	$4 < 5$, below threshold	Deactivates

5. Data availability

Approximately three months of measurement-partner digital-linear session data are available. You can find the corresponding datasets in the files you are provided.

Source Domain	File Name	Purpose
Digital-linear sessions	gfk_sessions_digital	Measurement-partner digital sessions used for activation logic.
Baseline synthetic sessions	tvi_spin_sessions	Existing synthetic linear-TV sessions generated by the platform.
Synthetic unit weights	spin_weights	Representational weights used for threshold and capacity rules.
Demographic mapping	virtual_target_groups	Mappings from detailed audience definitions to the six buckets.

6. KPIs

To verify that your solution is working as intended, we recommend comparing the generated synthetic output against the digital input at different aggregation levels (overall, per channel, demographic bucket, and time interval).

Suggested KPIs include:

- **Total Time Watched (TTW):** Compare weighted synthetic TTW against unweighted digital TTW.
- **Concurrent Weighted Viewers:** Compare the sum of active synthetic viewer weights with the number of active digital sessions over time.
- **Assignment Coverage:** Report the percentage of successfully assigned digital sessions and any rejected or unassigned sessions.
- **Demographic Consistency:** Verify that assignments remain within the correct demographic bucket.
- **Viewer Integrity:** Ensure that no synthetic viewer watches more than one channel at the same time and that activation, deactivation, and capacity rules are respected.

When presenting your solution, briefly discuss how well your output reproduces the input KPIs and explain any remaining deviations.

7. Your Task and the Expected Final Output

- The final output consists of session-shaped records structurally equivalent to the existing synthetic linear-TV sessions.
- Records must preserve channel, broadcast day, time interval, demographic bucket, synthetic unit identifier, and relevant representational weight fields needed for KPI calculation.
- No synthetic viewer is allowed to be watching on more than one channel at a time

Please design and implement a production-ready and runnable approach for this feature. You may implement it in SQL, Python or Rust, and support your thinking by providing architecture diagrams, data models, or a combination of these. The goal is not to memorize platform internals; the goal is to reason about stateful assignment, weighting, correctness, and production safety.

Provide your solution using a GitHub Repository. Feel free to support your thoughts and decisions regarding algorithmic design, data modelling, performance decisions and more by providing a README. Also ensure the correctness by thinking about testability or reproducibility.